

Review:

A Tutorial in Dynamic Systems Simulation and
Modelling With R, *ctsem*, and *lme4*: Couples'
Affect Dynamics Over Time
(Manuscript id: MET-2026-0505)

This tutorial paper explains how to build a dynamic model and how to estimate versions of such model with *ctsem* package. I think we need a lot more thoughtful iterative model building and more accessible resources on this topic. This tutorial both shows how to do that for a simple example and also showcases the capabilities of *ctsem*, both of which are useful. I have no technical concerns. However, while reading the paper I tried to collect some didactic points which might help the authors to further strengthen their tutorial so that it is more helpful for their target audience. I have a couple of larger points, and then a lot of smaller points below.

Major Points

1. **Explain general rationale more at the outset:** The introduction says that you will walk the reader through iterative model building. However, it is perhaps not obvious to everyone that this means that you will create generative models based on theory and then recover these models from synthetic data. It is not really explained why this is happening. I think this would be good to lay out in some more detail, so that the reader can better appreciate the material that follows. Now this is left implicit, and a reader with little background in modeling might be wondering why you are not just fitting successively more complex models.
2. **Jump in difficulty when introducing *ctsem*:** I think there is a big jump in the difficulty of the material when transitioning to the section on *ctsem*. Before things are still quite easy to follow and you still provide all the code. But on page 11 the difficulty greatly increases. New ideas are introduced much more quickly, and I don't feel anymore taken by the hand and guided through everything like in the first 10 pages of the manuscript. For example, you just mention the latent dynamic model

in passing, without ever really motivating or discussing it in the context of the running example. Figure 18 is also not really explained in great detail, which might be difficult to process especially for the less technically inclined readers, who I suspect are your target audience. This is handled better in Figure 14 though, where there is more of a walk-through.

In any case, I felt like the difficulty level was increasing quite a bit. This might be addressed by explaining a bit more here and there. But also perhaps by warning the reader that this will happen.

3. **Paper Length:** The paper is pretty long and you have 32 (!) figures in the main text. As someone who is reasonably knowledgeable of what you are presented here and who found the paper a very interesting read, I found it challenging to read it in one sitting. I could imagine that it would take readers with less of a technical background even more resolve. This can be fine if you see the paper more of a handbook where one can look things up. But if the idea is that people read the whole paper front to back, you could consider shortening the material a bit. One option could be to only sketch the later models in the main text and move the details into the appendix.
4. **Possible Models and Behavior:** I was wondering whether you could comment a bit more in the end on the possibility of nonlinear models. For example, can the latent processes interact? That of course would allow one to model systems with much more complex behaviors than the ones in the running example, which I think were all single fixed point systems. Related, estimating initial states and using the (V)AR(1) transient to get a nonlinear effect towards the single fixed point is a great use of this model class. However, this of course does not provide a model for how a person ended up in their initial state in the first place. This is perhaps another higher-level topic to comment on in the end.
5. **Data Requirements:** Since the paper opens very generally with dynamical systems, perhaps you could also comment on the data requirements of ctsem. Clearly the complexity of models must be limited by the amount of data that is available.

In addition, it would be interesting to hear the author's big picture view on creating dynamical systems models in psychology. Specifically, do you think that we can always fit the whole model to a single dataset like you do here? If not, what are alternative approaches? How could one in principle combine different sources of information to recover a dynamical system? Of course, I don't expect you to solve any of these issues, but a (brief) discussion of these questions could help relate your approach to other approaches.

Minor Points

1. Small thing: On page 2 in the third list point you conceptualize noise as "reflecting unpredictable events", which almost makes this sound like a limitation. But I guess these are just the causal effects on the modeled variables that we choose to abstract away with a probability distribution?
2. Page 2: "our first idea"; I don't follow why this would be the simplest idea you can come up with. I would think the simplest idea is that nothing changes.
3. page 3: I found it very curious that you conceptualize $\epsilon(t)$ as measurement error. Is this really the most interesting stochasticity to model here?
4. page 3: I was missing a comment on the (measurement) scale of affect, and the choices of the parameters (trend, noise variance). Of course, I understand that you want to keep this concise, but if you go for this first-principle tutorial (which I think is great), it feels odd to just introduce these unmotivated parameter values.
5. Page 4: you show the correlation between random effects to be -0.92. It is not entirely clear from the text before why we would expect that.
6. Page 4: in the model with 20 subjects, the measurement error variance is 0.1, while above for the single person it is 1. Why?
7. Figure 2: Maybe you can color the people along their intercept values, to make this correlation between the random effects visually more obvious
8. On page 6 you say "The solid line does not map well onto the observed trajectory of affect, and there is a large amount of uncertainty about the model-implied trajectory." perhaps you can also briefly comment on this "mismatch" (regularization due to the empirical bayes of the multilevel model). But I'm not sure anymore what is happening here (see next comment).
9. Page 6: I did not understand what is shown in Figure 4, which I think you refer to with "the second plot".
10. Page 6: You write "By formalizing our linear growth process for Jill and other patients, we observe that our simple conceptual idea leads to a surprising prediction. Namely, therapy actually helps people with a lower initial level of affect to have a higher affect level at the end of the 21 weeks of therapy, compared to people who started with higher affect." I find it a bit odd to frame this as a discovery of the analysis, because this is the model we specified and simulated from.

11. On pages 6-7 you introduce a non-linear effect via the transient of an AR(1) process. I was a bit confused by this, because before you modeled the linear effect with a deterministic linear trend. So for me the logical step would have been to model this with a non-linear deterministic trend. Not saying that this is what you should do in practice, but that would have followed more logically. Especially, since you do not announce this conceptual switch. I think it would be good to discuss this change in the modeling approach more explicitly.

Perhaps also comment on the problem that this is perhaps a bit of an odd model for therapeutic change, because when you start *above* B, your affect will *decrease*.

12. In the first paragraph on page 8 you mention some advantages of CT over DT modeling. If you won't expand on this later, I think it would be good to do this somewhere. Currently, you can only appreciate these points if you already know about them.

After reading a bit more, I feel like it would be good to better motivate CT already here. Because you basically have to give a mini-calculus course here, and especially people who don't find this easy might not understand why they have to go through all this trouble, if they didn't really understand why CT is tangibly better than DT

13. Making the step from discrete time to continuous time could be softened by rewriting the DT model into a difference equation, and then say that the time step becomes arbitrarily small, giving us the derivative on the LHS
14. Linear ODE: Perhaps you can show the function η and then take the derivative to get B . Maybe this clicks better for people who ever had calculus and remember the power rule
15. Page 8: When switching to the simulation, now the start of "Simulation" is very technical. Perhaps you could explain a bit more which problem you have to solve here and how you solve it with this numerical approximation.
16. Page 8: Euler's method: I was surprised to see that you only take 20 time steps. Don't you usually need more to get a good approximation of the analytical solution (which here you could still derive)? It might also be confusing for the reader, because basically you are doing the same as in the DT case.

Okay, I now realize that you discuss this right after. Perhaps add a sentence saying that you will first approximate this and then make this more precise.

This could also be a place to cut material and make the paper shorter. That is, immediately add many steps.

17. Page 8: You here for the first time introduce stochasticity that is pushed through the system and that is not measurement error. I think it could be nice to discuss more what we are modeling with this, because I think conceptually this is a big step.
18. infinitesimal: I'm not a mathematician myself, but my understanding is that this has been replaced with the limit definition / and "arbitrarily small" (and not: infinitely small, which is not a real number)
19. I'm not sure whether "system noise" was properly defined at some point.
20. At the end of page 10, when introducing ctsem you mention "state-dependence" but it's not clear to me how this relates to the above. In general, the switch is pretty abrupt, because you introduce ctsem as a way to fit a latent dynamical model. But so far, we did not look at such models, so this came a bit out of nowhere for me.

Reading further on, I think it would be good not only to introduce the state-space latent-measurement model setup, but also motivate why this is useful in the case of the running example.
21. Figure 8: Perhaps it would be good to briefly talk the reader through the model structure again. This could also tie the earlier discussion of all pieces together.
22. Figure 10 and 11: Similarly to above, I am not sure why we are looking at this analysis.
23. Page 14: Jill continues "Consider Jill, who attends weekly therapy sessions during her 20-week treatment period. Each session provides a temporary boost to her affect. The longer term effect of the therapy is already accounted for by the growth trajectory we have already specified in earlier examples." This is where the running example for me finally departs from reality. I don't think that this is a fair representation of how any therapy works. Not sure how to deal with this. Perhaps just acknowledge it? I don't know.
24. Perhaps you can choose two neutral names to make the couple example less hetero-normative. Like Alex and Robin or something like that.
25. page 16: "Correlation in system noise, random life events that cause fluctuations in Jill's affect trajectory are also likely to contemporaneously impact Bert's affect (and vice versa)." I think it would be interesting for the reader to have spelled out why this is the case.
26. page 23-24: Adding time-dependent moderators: you give the dummy-coded variable indicator communication as an example. Can you also add interaction terms between the modeled state-variables? So, in your example, the cross-lagged effect of Jill's affect on Bert's affect depends also on Bert's current affect.

27. In the model selection you seem to use within-sample R^2 . However, wouldn't this R^2 only go up when estimating models that are too complex? When they are nested, they would technically not be misspecified, but of course you would never be able to select a data generating simpler/constrained model.

This is a signed review:

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